

Multiple Choice (10 marks)

1. Insulin deficiency is associated with
 - (a) Reduced lipolysis
 - (b) Increased ketogenesis
 - (c) Reduced gluconeogenesis
 - (d) Reduced proteolysis
2. What is the best method to estimate changes in body fat during weight loss?
 - (a) BMI
 - (b) DXA
 - (c) Four-compartment model
 - (d) MRI
3. From the list of oral microorganisms, which is primarily responsible for the initiation of dental caries?
 - (a) Mutans streptococci
 - (b) bifidobacteria
 - (c) Lactobacilli
 - (d) P. gingivalis
4. Which of the following inborn errors of metabolism gives rise to zinc deficiency?
 - (a) Acrodermatitis enteropathica
 - (b) Wilson's disease
 - (c) Menkes disease
 - (d) Haemochromatosis
5. The gut microbes metabolize L-carnitine into which metabolite that has been shown increase cardiovascular disease?
 - (a) choline
 - (b) betaine
 - (c) Trimethylamine
 - (d) All of the above

True / False (10 marks)

Answer True or False

6. True or False: Natural polyunsaturated fatty acids in vegetable oils usually have 18 carbon atoms with one carbon-carbon double bond in the cis configuration.
7. True or False: Fish is a common source of goitrogens that can interfere with thyroid hormone production.
8. True or False: Vitamin A is essential for the activation of vitamin D receptors in the body.
9. True or False: Gluten may increase carbohydrate malabsorption in some individuals.
10. True or False: A sedentary 43-year-old woman who weighs 63 kg requires 1800 kcal per day for her energy needs.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of niacin in fatty acid synthesis, focusing on how it functions as a coenzyme in reduction reactions and its importance in overall metabolic processes.
12. Discuss the various factors that contribute to the increased risk of vitamin and mineral deficiencies in older adults in high-income countries, excluding the role of high nutrient requirements for tissue turnover.

End of Paper